Stochastic Valuation of Pine Stands Using Monte Carlo Simulation

by

Charles E. Merritt IV

(Under the direction of Pete Bettinger)

Abstract

Stand-level forest planning is commonly conducted with deterministic growth and yield models that assume a single, idealized trajectory of stand development. However, forest stands evolve under substantial uncertainty arising from stochastic growth variation and disturbance events such as pests, wildfire, and wind damage. This thesis develops a modular stand simulation framework that augments an established deterministic loblolly pine (*Pinus taeda*) growth model with stochastic process noise and probabilistic disturbance regimes, and uses Monte Carlo simulation to quantify how uncertainty alters economic valuation.

In this work, we evaluate a set of predefined, interpretable management prescriptions under both deterministic and stochastic dynamics. For each prescription and risk regime, repeated simulations generate a distribution of discounted cash flows and corresponding Net Present Value (NPV), with Land Expectation Value (LEV) reported when appropriate to enable comparison with classical forestry benchmarks. Outcomes are summarized using measures of central tendency and dispersion as well as downside risk metrics, including Value at Risk (VaR), Conditional Value at Risk (CVaR) to demonstrate tail-risk.

Results show that deterministic valuation can systematically mischaracterize stand value under uncertainty, particularly when low-probability, high-severity disturbances are possible. Across increasing disturbance frequencies and severities, the stochastic valuation distributions exhibit substantial tail risk and increased variability relative to deterministic baselines,

and financially preferred rotation ages and prescriptions may shift accordingly. Collectively, these findings demonstrate the importance of distributional valuation in long-horizon timber management and provide a practical framework for comparing prescription robustness under stochastic disturbance regimes.

Index words: Forest Value, Markov Decision Processes, Forest Growth & Yield Modeling, Simulation, Digital Forestry, Theses (academic)

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B.A., The University of Georgia, 2024

A Thesis Submitted to the Graduate Faculty
of The University of Georgia in Partial Fulfillment
of the
Requirements for the Degree

Master of Science

Athens, Georgia

2026

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Dedication

I dedicate this work to my mother, whose unwavering support and guidance have shaped me into the person I am today. Your belief in me has always been my greatest strength. To my siblings, Collier, Caiden, and Annalynn, your eyes inspire me daily to set a worthy example and continuously motivate me to become a better version of myself. Thank you all for keeping me grounded and focused on what truly matters.

Acknowledgments

This work was supported by the U.S. Department of Agriculture, National Institute of Food and Agriculture, Sustainable Agricultural Systems program, PERSEUS grant, #2023-68012-38992. The findings and conclusions in this work have not been formally disseminated by the U. S. Department of Agriculture and should not be construed to represent any agency determination or policy. The U.S. Government is authorized to reproduce and distribute reprints for governmental purposes notwithstanding any copyright annotation therein.

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Chapter 1

Introduction

1.1 Background and Context

Forest management decisions often span long time horizons, necessitating methods that anticipate future conditions and trade-offs. Yet most current stand-level planning models are deterministic, assuming a single, idealized trajectory of forest development. These models typically use optimization techniques to identify regimes that maximize a static objective (or set of objectives), most often timber volume or forest structure goals, though stands may also be managed for carbon sequestration, wildlife diversity, or recreation. In doing so, they often overlook a critical reality: forest stands are not deterministic environments.

Rather, forest stands are subject to stochastic and dynamic processes. Growth trajectories are influenced not only by competition and ecological succession, but also by random disturbances such as wildfires, windthrow, ice storms, and pest outbreaks. These events are often abrupt, severe, and difficult to predict. Their effects propagate over time, altering successional pathways and disrupting static management plans. Despite this, such disturbances are often omitted from standard planning methods, which focus on idealized scenarios rather than risk-resilient strategies. Furthermore, disturbance events beyond a threshold of severity substantially increase the likelihood of sustained disturbances. For example, woody debris caused by wind damage increases forest fuel loads and subsequently the likelihood of wildfire while also increasing the susceptibility of trees to infestation from pests such as pine beetles due to their weakened state.

These random disturbance events are projected to become more frequent and severe. Recent history bears this out. For example, Georgia Forestry Commission (2024) reported

that Hurricane Helene caused $1.28 billion in damage to Georgia's forests, and affected up to 8.9 million acres of forest land in Georgia alone. Other events, such as the 2007 Georgia Bay Fire and the 2011 Honey Prairie Fire, burned 560,000 acres and 300,000 acres respectively. These events illustrate the vulnerability of forest systems to random, high-impact events in the Southern United States.

This growing uncertainty calls for a new modeling paradigm, one that explicitly represents stochastic risks and adapts to variable conditions over time. Rather than optimizing for a single, fixed trajectory, we propose a simulation-based framework that captures a range of plausible forest futures. While no simulator can perfectly reproduce the complexity of forest growth and disturbance, our intent is not to create a definitive predictive model for growth *and* stochastic disturbances in timber stands. Instead, the simulation environment serves as a modular testbed for evaluating management strategies and their effects on stand value under uncertainty. Its design allows for replacement or extension of the growth or disturbance models with site-specific or more sophisticated alternatives. By augmenting a deterministic growth and yield simulator with stochastic noise and disturbance events, we construct a flexible modeling environment in which multiple stochastic risk scenarios can be explored and evaluated against deterministic baselines.

To structure the stochastic elements in this environment, we model stand development as a probabilistic state transition process. At each annual timestep, stand attributes evolve according to established growth and yield equations, with additional variability introduced through stochastic process noise and disturbance events. This formulation allows us to generate multiple plausible realizations of stand development under identical management prescriptions.

Uncertainty in forest stand development can be categorized into two fundamental types. Aleatory uncertainty arises from inherent randomness in natural processes, including annual growth variability, weather fluctuations, micro-site heterogeneity and the stochastic occurrence of disturbance events such as fire, wind, or pest outbreaks. This form of uncertainty is

irreducible and must be propagated through simulation. Epistemic uncertainty, by contrast, stems from incomplete knowledge: imprecision in growth model parameters, limited calibration data, or structural assumptions about disturbance severity distributions. While epistemic uncertainty can in principle be reduced through additional data or improved models, it remains a practical constraint in operational forest planning. Our framework addresses both types: process noise and disturbance modules capture aleatory variability, while scenario-based analysis and sensitivity testing quantify the influence of epistemic factors on management outcomes.

Risk is evaluated using distributional financial metrics rather than single point estimates. For each prescription and disturbance regime, Monte Carlo simulation produces a distribution of discounted cash flows and corresponding Net Present Value (NPV). These distributions are summarized using measures of central tendency and dispersion, as well as downside risk metrics such as Value at Risk (VaR) and Conditional Value at Risk (CVaR). This approach allows comparison not only of expected returns but also of resilience to rare but severe disturbance events.

By reframing stand valuation as a distribution rather than a deterministic forecast, this framework provides a more transparent view of the economic consequences of uncertainty. In doing so, it bridges classical deterministic growth and yield modeling with modern risk analysis, while remaining grounded in established forestry finance principles such as NPV and LEV.

This thesis addresses several gaps:

- The ubiquitous tendency to sidestep uncertainty in deterministic optimization models which leave out many probabilistic events that are relevant to growth but complicated to model as described by Stevens (1986).

- The limited use of distributional valuation metrics in stand-level planning, where deterministic point estimates often mask downside risk.

- The missing link between tried and true empirical stand growth models (Harrison and Borders 1996) and stochastic models that include both chaotic disturbances and growth factors.

We propose a framework that embraces stochasticity from the outset by embedding probabilistic disturbance processes within an established growth and yield model and evaluating heuristic "rule of thumb" management prescriptions through Monte Carlo simulation. This approach enables direct comparison between deterministic and stochastic valuation outcomes under varying disturbance regimes.

This thesis investigates how probabilistic disturbance modeling can be integrated with established growth and yield models to evaluate forest stand value under uncertainty.

1.2 Stochastic vs. Deterministic Planning

In this work we address three key questions that challenge the use of deterministic modeling of pine stand development.

How does deterministic stand valuation differ from stochastic valuation under identical management prescriptions?

This question evaluates the extent to which conventional deterministic planning methods, such as fixed-rotation management optimized via LEV, diverge from valuation results obtained when stochastic disturbances and growth variability are explicitly modeled. Rather than comparing learned policies to deterministic schedules, we hold management prescriptions constant and examine how introducing stochastic disturbance processes alters expected value, variability, and downside risk.

How large is the bias introduced by ignoring disturbance risk and process uncertainty?

Deterministic stand models implicitly assume a single, idealized development trajectory. In contrast, stochastic Monte Carlo simulation generates a distribution of possible futures. This question assesses the difference between deterministic point estimates of value and

the mean, median, and tail outcomes of the corresponding stochastic distribution. Particular attention is given to downside risk, including the probability of substantial loss under elevated disturbance regimes.

Do changes in disturbance frequency and severity alter financially optimal rotation ages for even aged forests and prescription performance?

Classical optimal rotation theory identifies a single rotation age T^* under deterministic assumptions. Under stochastic disturbance regimes, however, realized rotation values are distributed rather than fixed. This question investigates how increasing disturbance frequency and severity shift the distribution of economically optimal rotation ages for even-aged forests and whether prescriptions derived under deterministic assumptions remain robust when evaluated across stochastic realizations. In particular, we examine whether higher disturbance risk systematically favors shorter rotations or alternative rule-based prescriptions.

1.3 Simulation and Valuation Framework

In this work we address two important questions related to ackowledging uncertainty in pine stand development.

How can a stochastic stand simulator be structured to evaluate management prescriptions under uncertainty?

We model stand dynamics as a stochastic state transition process governed by mechanistic growth equations and probabilistic disturbance events. Management actions are specified through interpretable, rule-based prescriptions such as fixed rotation ages, thinning thresholds, and salvage responses. Rather than solving for an optimal policy through reinforcement learning, we simulate stand trajectories under each prescription across many Monte Carlo realizations to estimate the distribution of financial outcomes.

Which valuation and risk measures are appropriate for comparing deterministic and stochastic outcomes?

For each simulated trajectory, we compute discounted cash flows to obtain Net Present

Value (NPV), and when appropriate, Land Expectation Value (LEV). Across Monte Carlo realizations, we summarize results using measures of central tendency and dispersion, as well as downside risk metrics such as Value at Risk (VaR) and Conditional Value at Risk (CVaR). These measures allow comparison not only of expected returns but also of resilience to catastrophic loss.

Together, these questions clarify how incorporating stochastic disturbance processes changes economic valuation and prescription robustness in stand-level forest management.

1.4 Modeling Assumptions

Every model has underlying assumptions, and it's worth discussing them early and directly. First and foremost, while the methods presented are general, almost all of this work was conducted with managed, even-aged, loblolly pine (*Pinus taeda*) plantations in mind. This simplifies the effort expended on factors such as species composition and competition. As such, our tree growth dynamics are modeled using region-specific equations provided by the Plantation Management Research Cooperative (PMRC). For the complete formulation of the deterministic growth model, see Harrison and Borders (1996).

The stochastic kernel parameters that dictate the probability distributions for disturbances and growth variability are scenario-based, not empirically estimated. We explore uncertainty via structured parameter scenarios that span plausible ranges. All stochastic elements (e.g., disturbance probabilities, process noise intensity) are stationary within a scenario. The distribution parameters are chosen at the beginning and held fixed through the episode, which is equivalent to the rotation.

We use a fixed rotation age of 35 years, slightly longer than the standard rotation for Southern pine plantations, to provide a consistent benchmark for comparing deterministic and stochastic outcomes. Discount rates, and management costs are also stationary and scenario-defined, they don't evolve stochastically within a simulation. Stationarity is an important mathematical assumption for Markov processes, and the goal of this work is not

to model the disturbance probabilities or economic factors themselves, but to understand the effect of these stochastic elements on stand value. For each of these values, we use the most recent estimates available from the literature.

Disturbances may have lagged effects (e.g., pest outbreaks causing gradual decline), which could be modeled with more complex recovery functions. In this work, disturbances do not alter future disturbance probabilities (e.g., one fire doesn't reduce or increase the chance of another, except indirectly through changed stand state). When disturbances occur, a salvage fraction of 50% is assumed available as revenue.

Stumpage and action prices are also stationary and defined by the latest literature Maggard and Natzke (2024). We impose the following feasibility constraints: TPA ≥ 100, BA ≥ 0, HD is non-decreasing, and Age > 0, where TPA is trees per acre, BA is basal area per acre, and HD is dominant tree height.

Thinning, when applied as part of a fixed heuristic policy, is assumed to be "from below" to remove trees from the smallest size classes first. In this investigation into uncertainty of pine stand development, we don't model harvest scheduling constraints, road access, policy restrictions, or multi-stand interactions, as this is framed as a single stand modeling problem.

Finally, while the stand transition process is Markovian in structure, this assumption serves as a modeling abstraction rather than a literal claim about ecological memorylessness. The state vector is constructed to capture sufficient information for future growth and disturbance response within the scope of this study. It is very likely that the true ecological process is not Markovian, but this is a common assumption and is a useful abstraction for the purposes of this work.

1.5 Thesis Statement

This thesis demonstrates that deterministic growth-and-yield projections systematically overstate expected stand value when stochastic disturbances are present. Under identical management regimes and a 35-year rotation, Monte Carlo simulations (n = 1,000) reveal

that even modest disturbance frequencies—a 30-year mean return interval with 30% mean severity—reduce mean Land Expectation Value (LEV) by 25% ($1,100/acre vs. $1,465/acre deterministic) and mean terminal harvest value by 21% ($5,156/acre vs. $6,553/acre). At higher but still realistic disturbance frequencies (10-year return interval), mean LEV declines by 63% to $545/acre, with 97% of trajectories falling below the deterministic benchmark. Process noise alone accounts for only a 5% LEV reduction, indicating that discrete disturbance events—not growth variability—drive the valuation gap. These findings suggest that forest investment analyses relying solely on deterministic projections may substantially overestimate expected returns and understate downside risk.

Chapter 2

Related Works

2.1 Theoretical Framework

This thesis develops a stochastic extension of a deterministic growth-and-yield model. The core modeling target is the distribution of terminal stand value under uncertainty:

$$p(V_T \mid s_0, \pi, \theta)$$

where V_T is terminal timber value, s_0 is the initial stand state, π is a fixed management policy, and θ parameterizes the stochastic transition kernel.

2.1.1 Stochastic Transition Structure

The stochastic transition factorizes as:

$$P(s_{t+1} \mid s_t, a_t) = P(z_t) \cdot P(q_t \mid z_t) \cdot P(s_{t+1} \mid s_t, a_t, z_t, q_t)$$

where $z_t \in \{0, 1\}$ indicates disturbance occurrence and $q_t \in [0, 1]$ is disturbance severity. s_t is the stand state at time t and a_t is the management action at time t. A deterministic growth model provides the baseline transition $\mu_{t+1} = f(s_t, a_t)$, interpreted as the mean biological trend. The stochastic extension introduces two sources of uncertainty:

1. **Process noise**: Multiplicative noise on growth increments, scaled by a global intensity parameter λ_{proc}. When this parameter is zero, the stochastic model reduces exactly to the deterministic baseline.

2. **Disturbance events**: Discrete shocks with random occurrence and severity, applied as proportional reductions to state variables.

This structure preserves the deterministic model as a limiting case while allowing systematic exploration of uncertainty regimes.

2.1.2 Relationship to Markov Decision Processes

While this framework can be viewed through the lens of Markov Decision Processes, several distinctions are important. First, the policy π is a fixed heuristic rule, not an optimized or learned policy. Second, this is a forward stochastic simulator, not a Bayesian inference procedure or reinforcement learning algorithm. Third, epistemic uncertainty over stochastic kernel parameters is represented through structured scenarios rather than posterior distributions, as empirical calibration data were unavailable.

The Markov assumption serves as a modeling abstraction: the state vector is constructed to capture sufficient information for future growth and disturbance response within the scope of this study, though the true ecological process may exhibit longer memory.

2.1.3 Simulation Target

For fixed initial conditions, policy, and scenario parameters, the object of interest is the distribution of terminal value estimated via Monte Carlo simulation. This allows comparison of expected values, downside risk measures, and the probability of falling below deterministic benchmarks across uncertainty regimes. Specific distributional forms, parameter values, and experimental design are detailed in the Methods, Chapter 3.

2.2 Current State of Knowledge

2.2.1 Net Present Value and Land Expectation Value

The standard economic approach to optimizing the set of management decisions for an even-aged stand of pine trees(equivalently the "management regime" or the "policy" in an RL context) is to first determine the value of a single rotation for a stand and a given rotation age based on Net Present Value (NPV).

The net present value (NPV) of harvesting a forest stand at rotation age T can be expressed as:

$$NPV(T) = \sum_{t=0}^{T} \frac{R_t - C_t}{(1+r)^t}$$

where:

- R_t is the revenue received at time t (e.g., from thinning or final harvest),
- C_t is the cost incurred at time t (e.g., planting, management, or treatment costs),
- r is the discount rate,
- T is the final rotation age.

Then, we expand on the definition of NPV to account for an infinite series of identical, even-aged rotations, starting from bare land. This is known as the Land Expectation Value (LEV). This formulation assumes that the same type of stand is regenerated and harvested infinitely many times on the same land. It can be expressed as:

$$LEV(T) = NPV(T) \cdot \frac{(1+r)^T}{(1+r)^T - 1}$$

where:

- $NPV(T)$ is the net present value of one rotation of length T,
- r is the discount rate,
- T is the rotation age.

The optimal rotation age T^* is defined as:

$$T^* = \arg\max_T \; LEV(T).$$

Once T^* has been defined for a stand, it is constant for the foreseeable future as long as business proceeds "as usual". Finding T^* is then a matter of some importance, it defines our "infinite and identical rotations" for the LEV. The standard method for establishing the optimal economic rotation age T^* is to use the LEV to compare multiple rotation ages until the process converges on the rotation age that returns the highest value. There are other methods for determining the optimal rotation age, for instance one could use the biological method. In that case, when the mean annual increment is maximized, it is time to cut the stand. Both methods yield viable choices for the optimal rotation age, and there are many other viable methods. (For a more detailed examination of methods relating to the assignment of T^* see Bettinger and Li 2009).

This method for calculating the value of a stand holds us to the assumption of "infinite and identical rotations at the optimal rotation age" in order to calculate the LEV. This assumption is extremely limiting, and in fact implicitly avoids uncertain outcomes. Nevertheless it remains an effective, easy, and robust way to calculate the economic value of a stand.

2.2.2 Stochastic Forest Disturbance Modeling

Accurate representation of forest disturbances is critical to keep our simulation as close as possible to the real dynamics of forest growth. To this end, we rely on the following references.

- "A hierarchical simulation-through-optimization approach to forest disturbance modeling" (Campbell and Dewhurst 2007). This work offers insights into a Markov approach for forest-level modeling. They very simply alter the standard harvest scheduling approach by treating disturbances as if they were another type of harvesting activity with no revenue or cost. In this way, disturbed stands are implicitly seen as having no value. This method is general enough to allow them to "model any stand-replacing disturbance..." regardless of age. In calculating the likelihood of these disturbances for their spatial model, firstly they retrieved historical data for their study area that had

similar rates of disturbances and harvest. Then, they discretize the likelihood of disturbance into decade-long bins, the same time-step used in their Markov model. Campbell & Dewhurst offer a comprehensive methodology for defining a spatial Markov chain for forest-level modeling, especially with regards to getting an implementation of disturbances themselves that is not too complicated.

- "Optimal Harvest Rates Considering the Risk of Forest Fire" (Stevens 1986). This thesis deals with a single risk type and offers some guidance on the ways in which people have tried to solve these problems historically. Specifically, it compares a traditional "even-flow" model with a dynamic programming model to compare how fire-rate impacts harvest volumes.

- "Dealing with Non-linearity and Uncertainty in Forest Management" (Messier et al. 2016). This article argues for a shift in forest management from traditional "command and control" approaches toward viewing forests as complex adaptive systems that prioritize resilience and adaptability. The authors propose incorporating non-linearity and uncertainty into practices by moving management targets from strict stand-level outcomes to a broader "envelope" of acceptable conditions.

- "Stochastic or deterministic single-tree models: is there any difference in growth predictions?" (Fortin and Langevin 2012). Critically, this paper shows that there are differences between stochastic and deterministic simulations due to interactions between "nonlinear mechanisms underlying forest growth."

- "Forest Growth and Yield Modeling" (Weiskittel et al. 2011). This textbook provides a comprehensive view of several major techniques for forest models, but also a sizable section on stand-level modeling as well.

2.2.3 Stand-Level Optimization Practices

Stand-level optimization in forestry begain in the 1970's using dynamic programming and simulation methods. The determination of the optimal rotation age of even-aged stands of trees as well as optimal thinning regimes were of initial interest. Later, optimizing stand density become an important consideration (Bettinger, Graetz, et al. 2005).

- "A density-dependent stand-level optimization approach for deriving management prescriptions for interior northwest (USA) landscapes" (Bettinger, Graetz, et al. 2005).

2.2.4 Markov Decision Processes in Forest Modeling

Markov Decision Processes (MDPs) provide a robust mathematical framework for modeling forest management under conditions of inherent ecological and economic risk. Unlike deterministic models that assume a known future, MDPs treat forest evolution as a series of stochastic transitions between discrete states. These states are typically defined by biophysical variables such as basal area, tree size, and species composition.

- "Managing Forests for Carbon and Timber: A Markov Decision Model of Uneven-aged Forest Management With Risk" (Johnston and Withey 2017). Focused on the Pacific Northwest, this study integrates carbon offset prices directly into an MDP objective function. It finds that while moderate carbon prices enhance species and size diversity, but high prices may inadvertently promote monocultures as the forest is managed primarily for carbon.

- "A Graph-based Markov Decision Process approach for managing forests under risk of wind damage" (Forsell et al. 2011). This article introduces Graph-based Markov Decision Processes to address the spatial complexities of wind damage risk in Swedish forests. By accounting for how neighboring stands provide topographical shelter, the model optimizes management policies to increase the expected NPV of stands.

- "Adaptive economic and ecological forest management under risk" (Buongiorno and Zhou 2015). This article presents an adaptive management framework for mixed stands in the Southern U.S., it highlights that ignoring stochastic shocks leads to inaccurate ecological predictions. Using the theory of MDPs, the authors develop a framework for managing forests under uncertainty according to economic and ecological objectives.

Chapter 3

Methods

3.1 Research Design

This study evaluates how deterministic and stochastic dynamics influence the economic valuation of a single forest stand under a fixed, rule-based management prescription. Rather than learning an optimal management policy under a continuous action-space (through a process such as reinforcement learning), we specify a single heuristic basal-area-triggered thinning rule and compare financial outcomes under deterministic and stochastic growth scenarios.

The simulation framework models stand dynamics through annual state transitions over a 35-year rotation. The stand state $s_t = (a_t, h_t, n_t, b_t, S, r, \phi)$ consists of four dynamic variables—age (a_t), dominant height (h_t), trees per acre (n_t), and basal area (b_t)—and three constants: site index at base age 25 (S), PMRC region (r), and percent hardwood (ϕ). Derived quantities such as volume and product yields are computed on-demand from these atomics. The stand is assumed to be even-aged and single-species (loblolly pine, *Pinus taeda*), consistent with plantation conditions across the non-coastal areas of the Southern United States.

Management is limited to a single heuristic thinning policy: thinning from below is triggered at age 15 when basal area exceeds 150 ft^2/ac, targeting a residual basal area of 100 ft^2/ac. Stumpage prices are fixed at \$9.51/ton (pulpwood), \$23.51/ton (chip-n-saw), and \$27.82/ton (sawtimber). Costs are also fixed: logging at \$150/acre, replanting at \$150.80/acre, and thinning at \$87.34/acre. Prices are derived from the latest TimberMarket

South report, while costs are derived from the Maggard and Natzke (2024) market report and online dashboard. A 5% annual real discount rate is applied for NPV and LEV calculations.

The stochastic extension introduces two sources of uncertainty layered onto the deterministic PMRC model (Harrison and Borders 1996):

- **Process noise**: Multiplicative lognormal noise on BA and HD increments, with mean correction ensuring $\mathbb{E}[\text{multiplier}] = 1$. TPA mortality uses a binomial model. A global scaling parameter $\lambda_{\text{proc}} \in \{0, 0.25, 0.5, 1.0\}$ controls noise intensity.

- **Disturbance events**: Annual Bernoulli occurrence with probability $p_{\text{dist}} \in \{0, 1/30, 1/20, 1/10\}$. Conditional severity follows $\text{Beta}(m_q\kappa, (1 - m_q)\kappa)$ with $m_q = 0.30$ and $\kappa = 12$. Shocks are proportional reductions to TPA and BA; dominant height is unaffected.

When $\lambda_{\text{proc}} = 0$ and $p_{\text{dist}} = 0$, the stochastic model reduces exactly to the deterministic PMRC model projection baseline. This design ensures that differences in financial outcomes arise from uncertainty rather than model discrepancies.

The simulation environment consists of three primary components: a growth module, a disturbance module, and an economics module.

3.2 Management Prescriptions

Management decisions are implemented through predefined, rule-based prescriptions rather than optimized control policies. Each prescription specifies treatment timing and intensity as a deterministic function of stand state variables. The prescriptions evaluated in this study include:

Fixed Rotation (FR-T)

A pine stand is harvested at a predetermined rotation age T. No intermediate treatments are applied unless explicitly specified. We evaluate a rotation age of 35 years to represent a plausible range for loblolly pine plantations owned by private forest landowners in the Southern United States.

Basal Area Threshold Thinning (BAT) A thinning from below is triggered when basal area exceeds a threshold BA^*. A fixed proportion τ of merchantable volume is removed at that time. Harvest occurs at a specified rotation age T.

Each prescription is applied identically across deterministic and stochastic simulations. This ensures that differences in financial outcomes arise from disturbance and growth variability rather than adaptive changes in management.

3.3 Growth Module

"Yield Prediction and Growth Projection for Site-Prepared Loblolly Pine Plantations in the Carolinas, Georgia, Alabama and Florida" (Harrison and Borders 1996) This work provides us with robust yield curves and growth equations that provide us with a basis for growth in our model.

The growth module projects ordinary stand development using equations from Harrison and Borders (1996). For a time step $\Delta t = 1$ year, state variables are updated in the following order:

$$a_{t+1} = a_t + \Delta t \tag{3.1}$$

$$h_{t+1} = f_h(a_t, h_t, a_{t+1}) \qquad \text{(Chapman-Richards)} \tag{3.2}$$

$$n_{t+1} = f_n(n_t, S, a_t, a_{t+1}) \qquad \text{(Survival equation)} \tag{3.3}$$

$$b_{t+1} = f_b(a_t, n_t, n_{t+1}, b_t, h_t, h_{t+1}, a_{t+1}, r) \tag{3.4}$$

This projection order respects the dependencies among state variables: age advances first, then height (which depends on age), then survival (which depends on site index and age), and finally basal area (which depends on all preceding variables).

The growth module is validated by comparing simulated trajectories to empirical growth patterns reported for managed loblolly pine plantations. Separating growth from disturbance dynamics allows clear interpretation of baseline stand development prior to stochastic modification.

3.4 Disturbance Module

Disturbances are modeled as a generic event with Bernoulli occurence and a conditional Beta severity drawn conditional on occurence. The severity is used to determine the proportion of merchantable volume lost by applying a one-time shock to the atomic variables of the stand state and updating the state accordingly, and then checking for feasibility.

The occurence of a disturbance in a given year is an independent Bernoulli trial with probability p_{dist}, where p_{dist} is defined as the mean return interval in years. This corresponds to a description of disturbances as "occuring once every n years" where n is the return interval.

Table 3.1: Disturbance probability by return interval

Return Interval	p_{dist}
10 years	0.10
20 years	0.05
30 years	0.033
None	0.00

3.5 Economics and Cashflow Modeling

The economics module converts simulated harvest and treatment decisions into annual cashflows. Revenues are calculated from harvested product classes (pulpwood, chip-and-saw, sawtimber) using fixed stumpage prices. Treatment costs are assigned according to regional estimates (Maggard and Natzke 2024).

For each year t, net cashflow is defined as

$$CF_t = \text{Revenue}_t - \text{Cost}_t.$$

If a disturbance occurs, a salvage fraction of 50% of damaged merchantable volume is assumed recoverable as revenue.

Financial performance over a rotation of length T is evaluated using Net Present Value (NPV):

$$NPV = \sum_{t=0}^{T} \frac{CF_t}{(1+r)^t},$$

where r is the discount rate. When appropriate, Land Expectation Value (LEV) is derived from the rotation-level NPV to enable comparison with classical deterministic forestry benchmarks.

3.6 Evaluation Metrics

For each management prescription and disturbance regime, Monte Carlo simulation produces a distribution of NPV outcomes. Results are summarized using:

- Mean and median NPV,
- Standard deviation,
- Value at Risk (VaR),
- Conditional Value at Risk (CVaR),

Deterministic valuation is obtained by running the same prescription with disturbance processes disabled. The difference between deterministic NPV and the Monte Carlo mean provides an estimate of bias introduced by ignoring stochastic risk.

This design allows direct comparison between deterministic planning assumptions and stochastic valuation outcomes, highlighting how disturbance frequency and severity alter expected returns and downside exposure.

Chapter 4

Results

For analysis of the stochastic model, we compare deterministic and stochastic trajectories under a shared management rule. Unless otherwise noted, each stochastic scenario is based on $n = 1000$ Monte Carlo trajectories over a 35-year planning horizon. Thinning is applied as a deterministic trigger at age 15 when basal area exceeds 150 ft^2/ac, targeting a residual basal area of 100 ft^2/ac.

Increasing disturbance frequency lowers dominant height and basal area relative to the deterministic baseline, as expected under repeated disturbance events (Figure 4.1). At the same time, trees per acre (TPA) may remain elevated under high-risk settings. Disturbances reduce standing stock, but non-severe years still admit positive recruitment into the smallest size classes. The combined effect can produce higher mean TPA even when stand structure and merchantable productivity decline.

Table 4.1 defines the scenario naming convention used throughout the results.

Figure 4.4 summarizes NPV distributions across the full factorial of process noise intensity (λ_{proc}) and disturbance probability (p_{dist}). As either source of uncertainty increases, median NPV declines and interquartile range widens. The deterministic baseline (dashed line) lies above nearly all stochastic medians, confirming that ignoring uncertainty systematically overstates expected returns. Figure 4.2 shows several validation metrics for the deterministic scenario in terms of stand development and product distributions. Figure 4.3 shows the manner in which thinning response is handled in the model using a competition index. Figure 4.5 presents mean NPV as a heatmap over the same parameter grid. The gradient

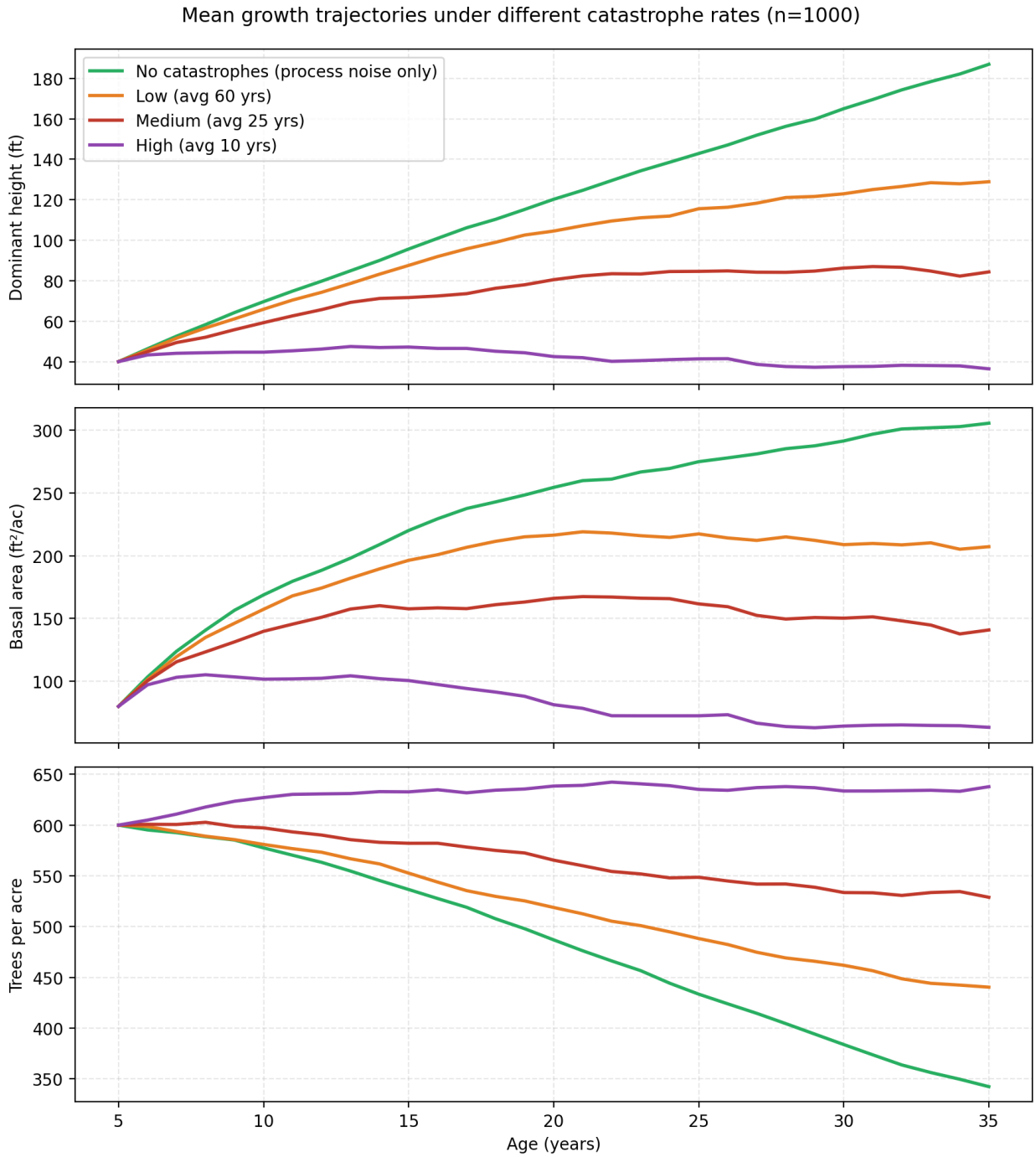

Figure 4.1: Mean dominant height, basal area, and TPA by disturbance regime ($n = 1000$ trajectories per regime).

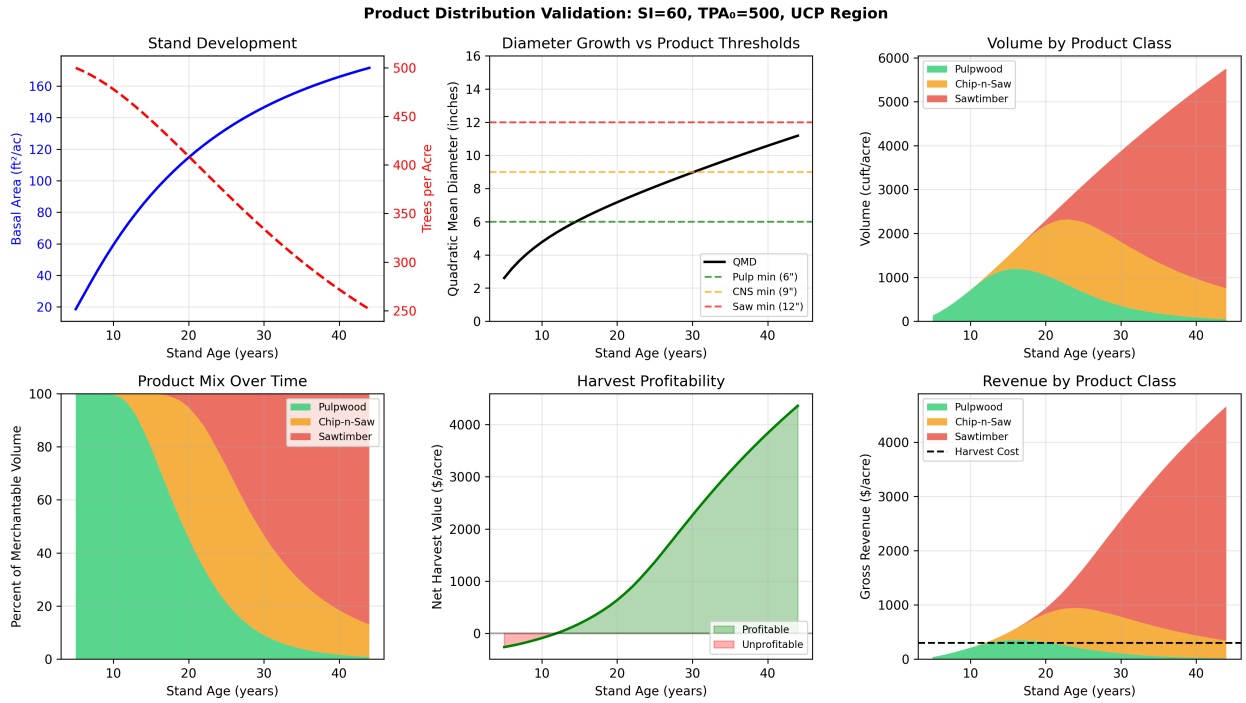

Figure 4.2: Product distribution following extended rotation (40 years).

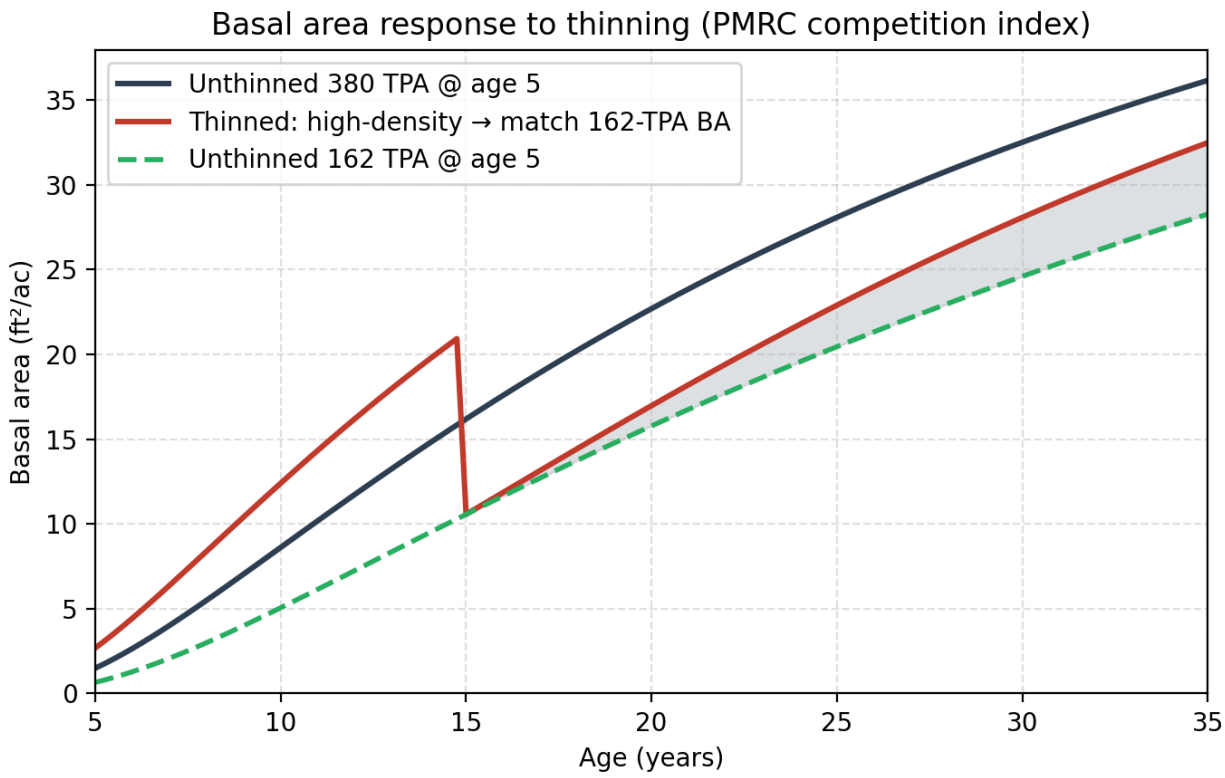

Figure 4.3: Deterministic thinning response relative to a 300-TPA unthinned reference baseline.

Table 4.1: Scenario naming convention

Code	Description	λ_{proc}	p_{dist}
`deterministic`	Deterministic baseline	0.00	0.00
`dist_30`	Disturbance only, 30-yr return	0.00	1/30
`dist_20`	Disturbance only, 20-yr return	0.00	1/20
`dist_10`	Disturbance only, 10-yr return	0.00	1/10
`noise_025`	Process noise only ($\lambda = 0.25$)	0.25	0.00
`n025_d30`	Low noise + 30-yr disturbance	0.25	1/30
`n025_d20`	Low noise + 20-yr disturbance	0.25	1/20
`n025_d10`	Low noise + 10-yr disturbance	0.25	1/10
`noise_050`	Process noise only ($\lambda = 0.50$)	0.50	0.00
`n050_d30`	Moderate noise + 30-yr disturbance	0.50	1/30
`n050_d20`	Moderate noise + 20-yr disturbance	0.50	1/20
`n050_d10`	Moderate noise + 10-yr disturbance	0.50	1/10
`noise_100`	Process noise only ($\lambda = 1.00$)	1.00	0.00
`n100_d30`	Full noise + 30-yr disturbance	1.00	1/30
`n100_d20`	Full noise + 20-yr disturbance	1.00	1/20
`n100_d10`	Full noise + 10-yr disturbance	1.00	1/10

from upper-left (low noise, low disturbance) to lower-right (high noise, high disturbance) illustrates the compounding effect of both uncertainty sources on expected stand value.

Figure 4.5 presents mean NPV as a heatmap over the same parameter grid. The gradient from upper-left (low noise, low disturbance) to lower-right (high noise, high disturbance) illustrates the compounding effect of both uncertainty sources on expected stand value.

Terminal harvest value distributions (Figure 4.6) show a similar pattern. Under high disturbance frequency, the distribution becomes increasingly left-skewed as severe disturbance events truncate stand development before merchantable volume accumulates.

The shape of the terminal value distribution shifts with disturbance regime. Figure 4.7 shows the distribution under moderate uncertainty ($\lambda_{\text{proc}} = 0.50$, $p_{\text{dist}} = 1/20$), while Figure 4.8 shows the distribution under elevated disturbance risk ($\lambda_{\text{proc}} = 0.50$, $p_{\text{dist}} = 1/10$). The

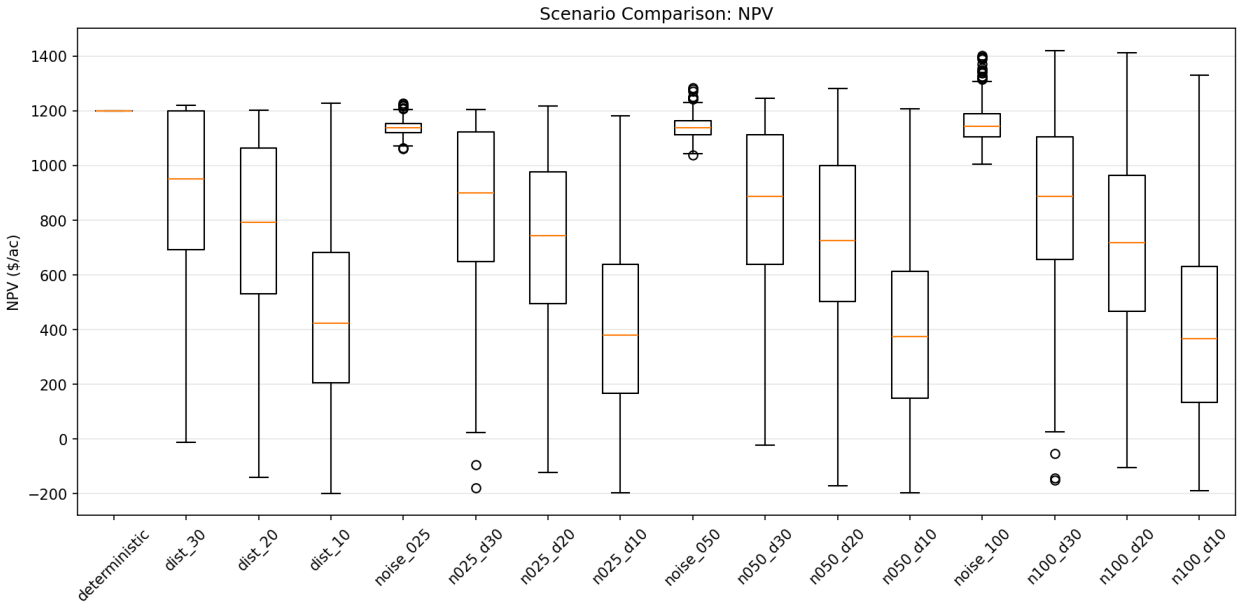

Figure 4.4: NPV distributions across scenarios defined by process noise intensity and disturbance probability ($n = 1000$ trajectories per scenario). Dashed line indicates deterministic baseline.

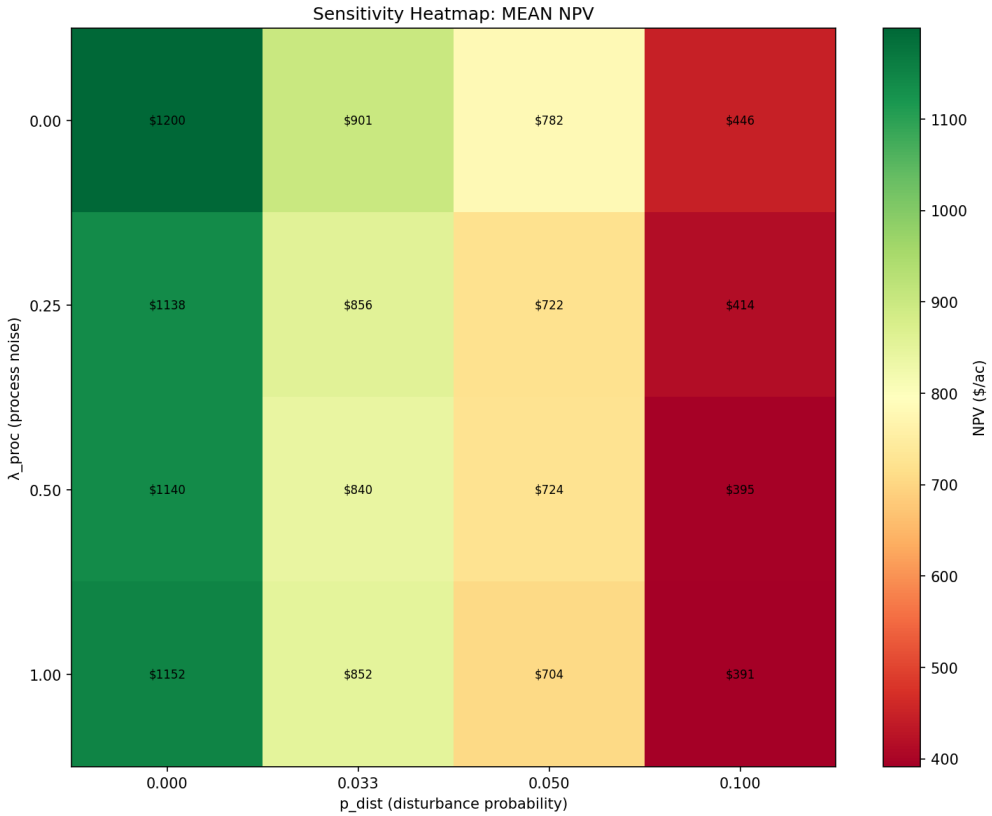

Figure 4.5: Mean NPV heatmap across process noise intensity (λ_{proc}) and disturbance probability (p_{dist}).

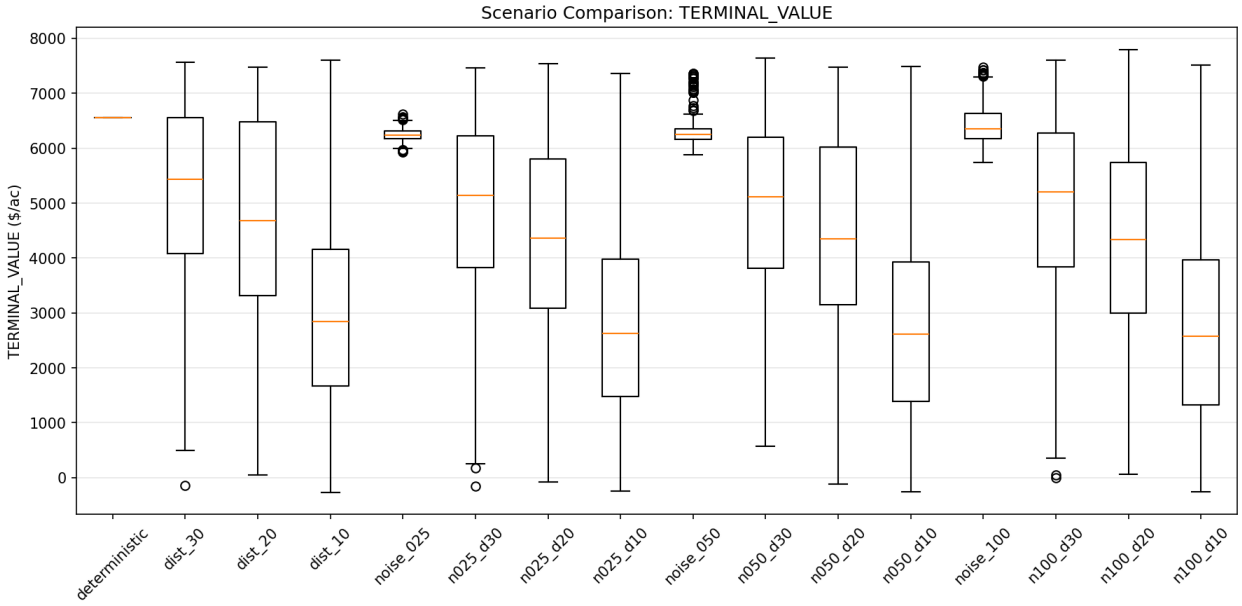

Figure 4.6: Terminal harvest value distributions across scenarios ($n = 1000$ trajectories per scenario).

latter exhibits greater left-skew and a heavier lower tail, reflecting increased probability of substantial value loss.

The full set of terminal value histograms for all stochastic scenarios is provided in the appendix.

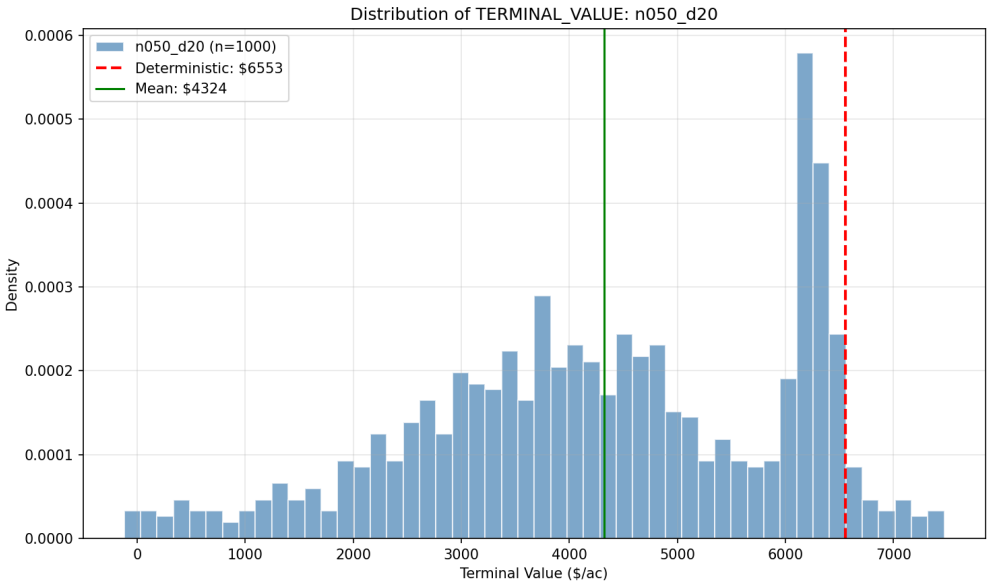

Figure 4.7: Terminal value distribution under moderate uncertainty ($\lambda_{\text{proc}} = 0.50$, $p_{\text{dist}} = 1/20$, $n = 1000$).

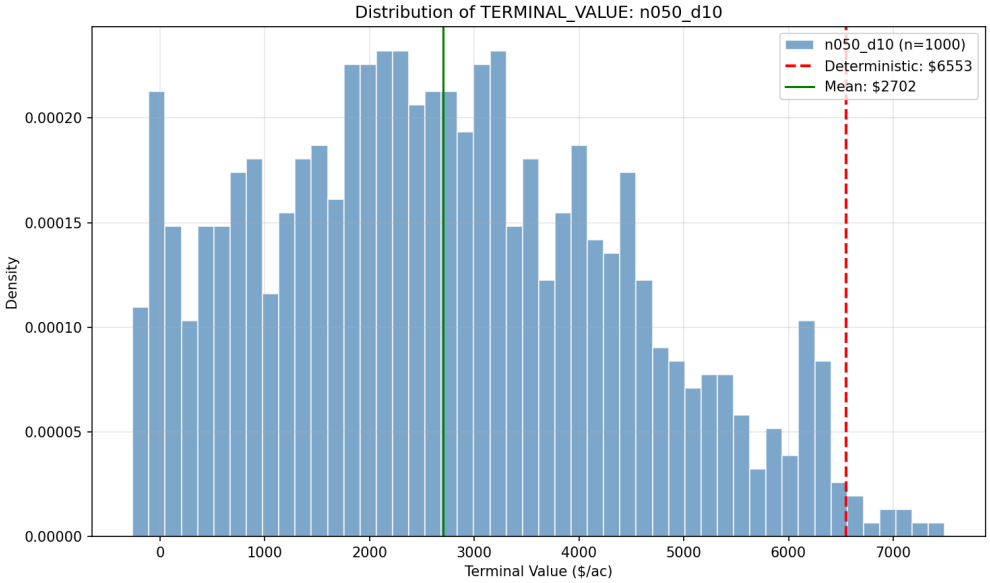

Figure 4.8: Terminal value distribution under elevated disturbance risk ($\lambda_{\text{proc}} = 0.50$, $p_{\text{dist}} = 1/10$, $n = 1000$).

Chapter 5

Discussion

The central result of this study is that disturbance risk changes both expected value and the shape of the value distribution of a pine plantation managed in the southern United States. Deterministic projections of growth and yield provide a useful reference point, but they systematically underrepresent downside dispersion when disturbance frequency increases. As risk increases, reductions in mean dominant tree height and basal area are accompanied by lower terminal stand value and wider value dispersion, indicating that expected-value comparisons alone are insufficient for management decisions in uncertain settings.

The trajectory diagnostics provide additional insight into the structural mechanisms underlying this valuation gap. Under increasing frequency of disturbances, mean dominant tree height and basal area diverge substantially from the deterministic baseline as repeated disturbances interrupt stand development. At the same time, tree density may remain elevated under high-risk scenarios due to repeated stand resets followed by regeneration and replanting at high initial densities. This creates a structural pattern in which stands maintain density but fail to accumulate merchantable volume. The resulting stand structure differs fundamentally from deterministic projections, which assume uninterrupted growth and gradual density decline through competition and thinning.

The systematic difference between deterministic and stochastic valuation arises from the asymmetric effects of disturbance on stand development. Growth variability alone introduces dispersion around expected trajectories, but severe disturbances impose irreversible structural shifts that damage accumulated merchantable volume. Deterministic simulations implicitly assume uninterrupted growth, thereby overweighting high-value trajectories that

are rarely realized when disturbance risk is present. In this context, the distribution of outcomes is the relevant decision object. Monte Carlo simulation does not replace silvicultural judgment; it exposes the range of plausible economic outcomes under modeled uncertainty. This improves transparency around downside exposure and avoids downplaying risk from a single-path forecast.

These findings are consistent with previous research showing that stochastic disturbances alter optimal management outcomes relative to deterministic projections. For example, Stevens (1986) demonstrated that incorporating fire risk into harvest scheduling can substantially change optimal harvest rates. Similarly, Fortin and Langevin (2012) showed that stochastic growth models produced different trajectories than deterministic growth models due to nonlinear interactions in the underlying forest growth dynamics. The present study extends this insight to stand-level financial valuation by demonstrating that these stochastic effects propagate directly into the distribution of economic outcomes.

This study intentionally uses rule-based management triggers rather than the full continuous action space of possible management actions. That choice keeps the management logic interpretable and operationally constrained, and supports direct deterministic-vs-stochastic comparisons under fixed prescriptions. In this setting, trigger rules (such as explicit thinning conditions) provide an auditable bridge between simulation output and practical forestry decisions. Furthermore, the vast majority of possible actions are not operationally feasible, and there is little information gain from exploring the full action space. This does not imply that continuous or high-dimensional action spaces are unimportant. Rather, for the present objective (valuation under uncertainty), rule-based triggers offer a reduced-complexity alternative while preserving explanatory power. Future work can evaluate when the additional flexibility of learned policies yields materially better outcomes.

From a management perspective, these results suggest that disturbance risk should influence both valuation methods and prescription selection. Under elevated disturbance regimes, prescriptions designed under deterministic assumptions may implicitly rely on long unin-

terrupted growth periods that are unlikely to occur in practice. Managers operating in disturbance-prone regions may therefore prefer prescriptions that capture value earlier in the rotation to reduce exposure to catastrophic loss. Although identifying optimal adaptive policies is outside the scope of this study, the distributional valuation framework developed here provides a basis for evaluating the robustness of management strategies under uncertain disturbance regimes.

Future work could include separate models for different types of disturbances, such as fire and wind, and different regeneration assumptions for different types of disturbances. Future work could also explore non-stationary disturbance processes, such as those that are driven by climate change. Similarly, the economic assumptions of the model could be modeled more extensively, perhaps by incorporating timber supply modeling. (Dehghan-Bonari et al. 2025).

Finally, future endeavors such as these might integrate and benchmark against broader ecological simulation frameworks. Pabst et al. (2008) provide concrete directions for extending disturbance and regeneration representation. A second extension might be to evaluate whether results are robust when uncertainty is embedded natively in growth-process formulations rather than layered onto deterministic projections. This approach reflects a broader trend in forestry research toward modular, high-dimensional models designed to accommodate diverse management objectives, such as LANDIS-II.

Chapter 6

Conclusions

By extending empirical, deterministic growth models with stochastic process noise and probabilistic disturbance regimes, a stochastic simulator was created to capture the uncertainties latent in the development of pine stands. This simulator was then used to evaluate the hidden cost of deterministic management policies and the importance of accounting for uncertainty in long-horizon timber management, even in the relatively short rotation age of the pine stands considered in this study. Deterministic valuation is therefore a limiting case in which transition variance is suppressed.

Deterministic simulations systematically overstate projected stand value relative to the true expected value under stochastic transitions, particularly in moderate-to-high disturbance regimes. This bias is driven by asymmetric downside effects: disturbances impose irreversible growth losses that are not offset by comparable gains. As uncertainty increases, the expected net present value declines and the variance of outcomes expands. In the highest risk regime examined, repeated severe disturbances can drive expected value negative under the assumed discount structure. The central findings are not only that expected value changes, but also that the distribution of outcomes widens and tail risk becomes economically meaningful.

Within the stochastic framework, disturbance risk alters transition probabilities and terminal reward distributions. Even under fixed management rules, valuation depends on the full transition kernel rather than a single trajectory. The expectation alone does not capture the structure of the decision problem. Monte Carlo simulation provides a tractable means of

estimating the value distribution when analytical characterization is infeasible under realistic dynamics.

For forest managers, these results imply that deterministic projections may remain operationally acceptable under low disturbance risk, but become increasingly misleading as risk intensifies. Ignoring stochasticity conceals downside exposure and produces optimistically biased valuations. Evaluating the full distribution of stand values enables calculation of risk metrics unavailable in deterministic frameworks and supports more transparent, risk-aware decision-making.

This work is a proof of concept and there are many directions for future work. Disturbance processes are stationary within scenario, recovery dynamics are stylized, and growth stochasticity is externally imposed rather than generated endogenously. Future research can proceed along three directions. First, model realism can be enhanced through non-stationary risk processes, richer disturbance structures, alternative regeneration assumptions, and spatially explicit stand representation. Second, growth models that are fundamentally stochastic, such as gap models from the ecological literature (Pabst et al. 2008) can be integrated to embed uncertainty at the process level. Third, the decision framework can be extended to include risk-sensitive objectives and policy optimization through reinforcement learning, allowing management strategies to adapt directly to stochastic environments.

By integrating standard forestry practices with stochastic simulation, we can evaluate the full distribution of stand values under uncertainty, and support more transparent, risk-aware decision-making in the critical domain of forest management. Taken together, these findings reinforce a central implication: once disturbance risk is introduced, stand valuation becomes inherently distributional. The remaining task is to formalize this shift and clarify its consequences for long-horizon forest management decisions.

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Appendix

Terminal Value Histograms by Scenario

This appendix presents terminal harvest value distributions for all 15 stochastic scenarios evaluated in this study. Each histogram shows the distribution of terminal values across $n = 1000$ Monte Carlo trajectories.

.1 Disturbance-Only Scenarios

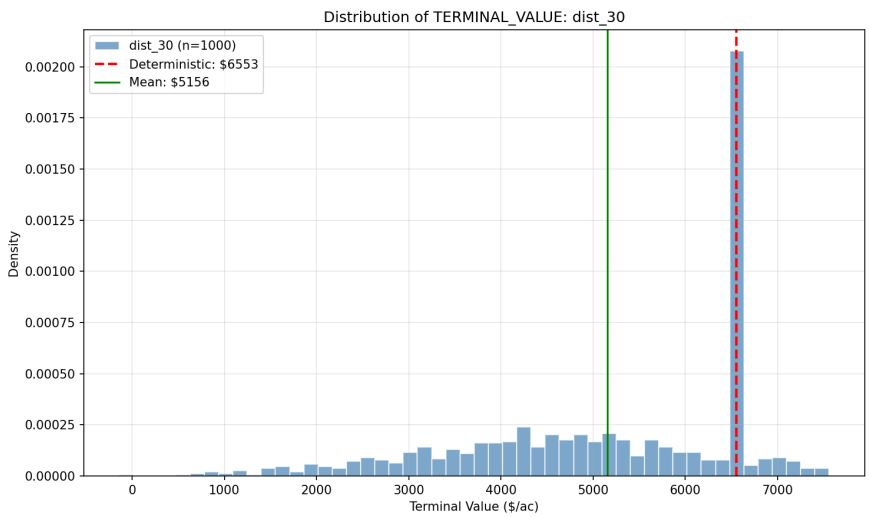

Figure 1: Terminal value distribution: `dist_30` ($\lambda_{\text{proc}} = 0$, $p_{\text{dist}} = 1/30$).

.2 Process Noise-Only Scenarios

.3 Combined Scenarios: Low Noise ($\lambda_{\text{PROC}} = 0.25$)

.4 Combined Scenarios: Moderate Noise ($\lambda_{\text{PROC}} = 0.50$)

.5 Combined Scenarios: Full Noise ($\lambda_{\text{PROC}} = 1.00$)

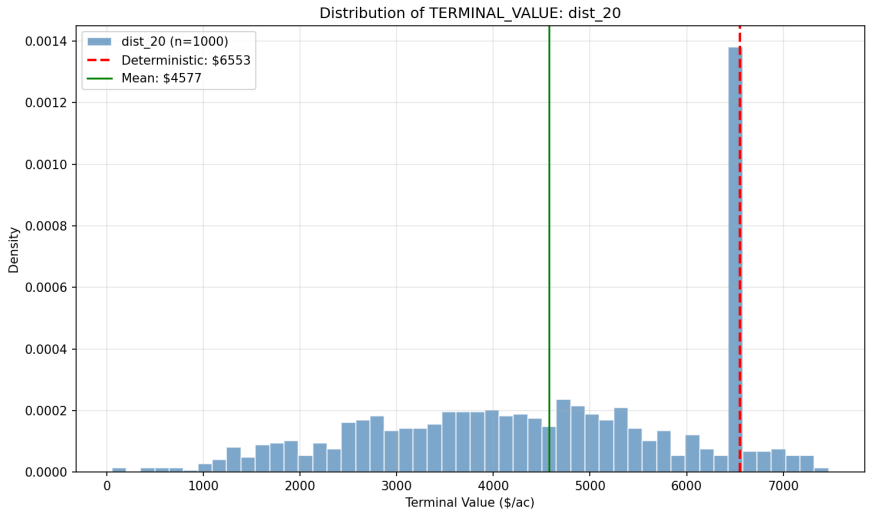

Figure 2: Terminal value distribution: `dist_20` ($\lambda_{\text{proc}} = 0$, $p_{\text{dist}} = 1/20$).

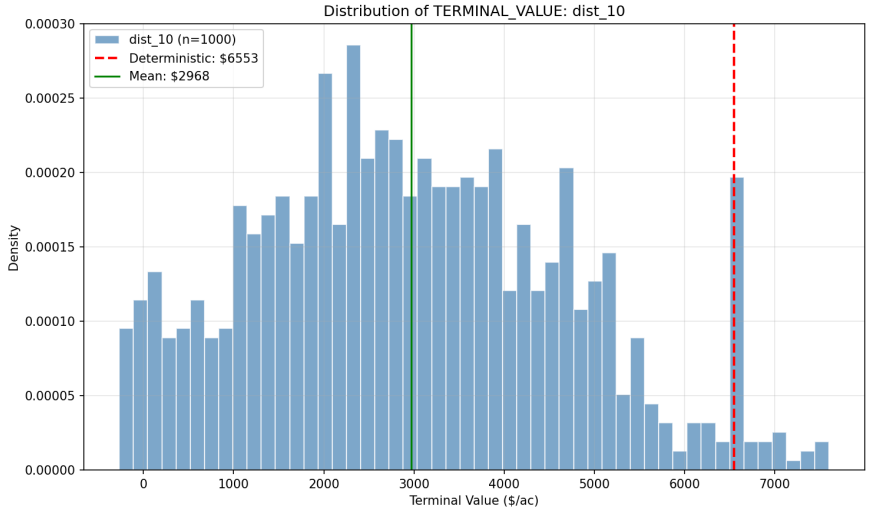

Figure 3: Terminal value distribution: `dist_10` ($\lambda_{\text{proc}} = 0$, $p_{\text{dist}} = 1/10$).

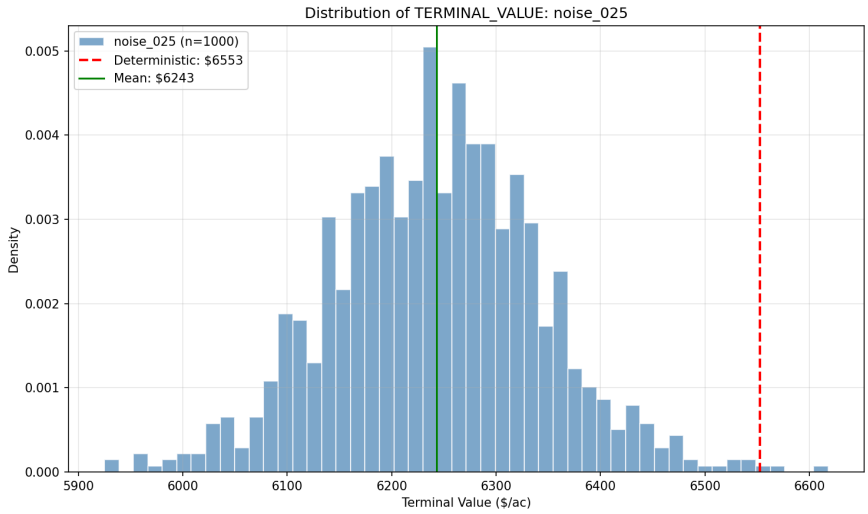

Figure 4: Terminal value distribution: `noise_025` ($\lambda_{\text{proc}} = 0.25$, $p_{\text{dist}} = 0$).

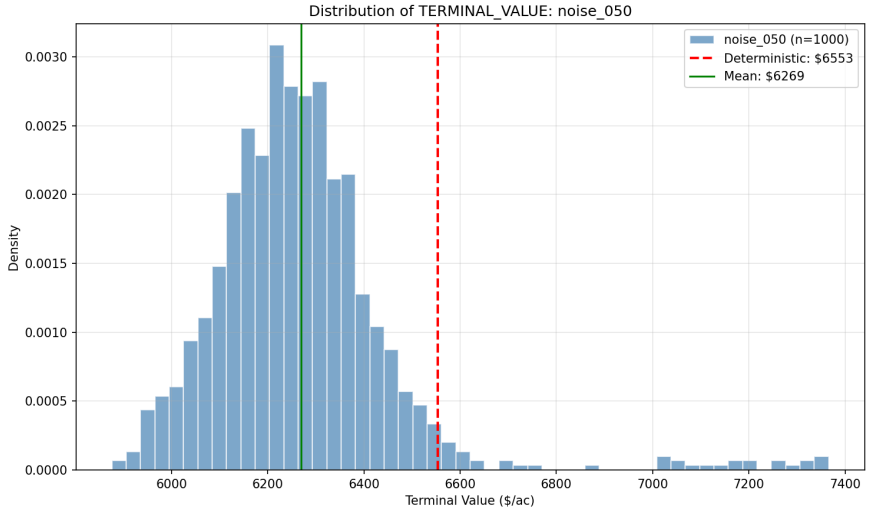

Figure 5: Terminal value distribution: `noise_050` ($\lambda_{\text{proc}} = 0.50$, $p_{\text{dist}} = 0$).

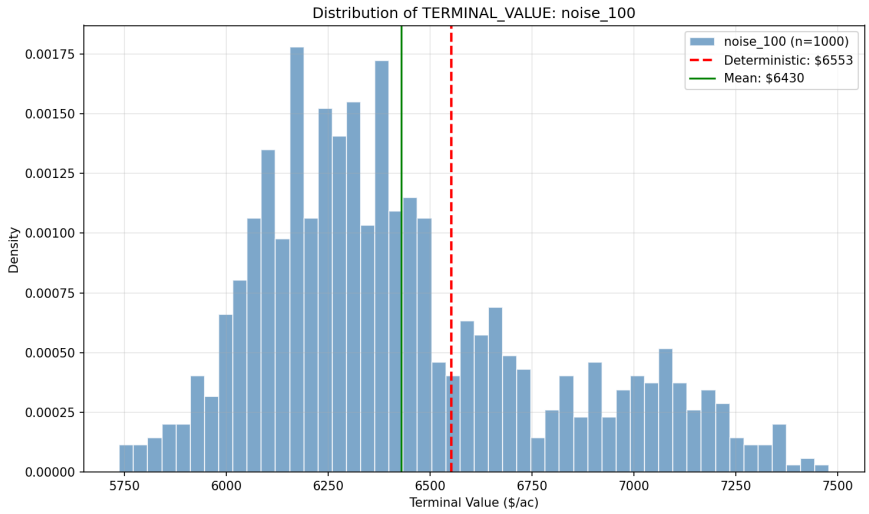

Figure 6: Terminal value distribution: `noise_100` ($\lambda_{\text{proc}} = 1.00$, $p_{\text{dist}} = 0$).

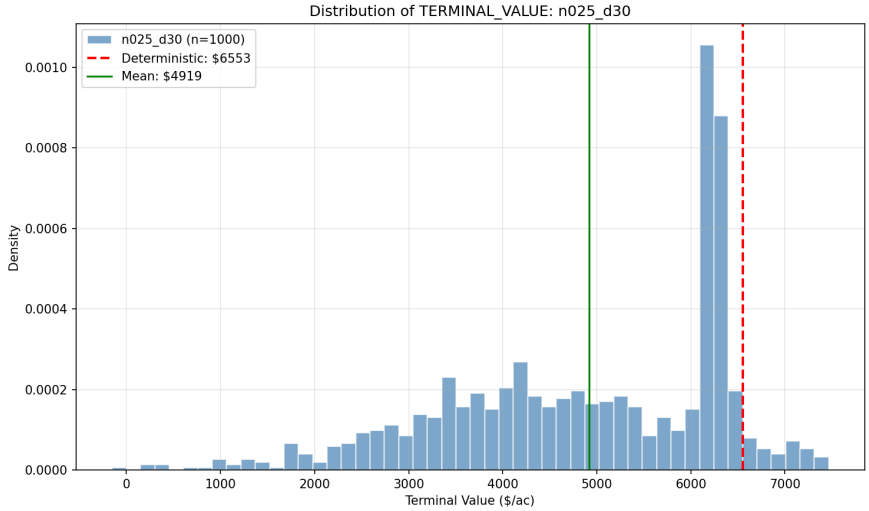

Figure 7: Terminal value distribution: `n025_d30` ($\lambda_{\text{proc}} = 0.25$, $p_{\text{dist}} = 1/30$).

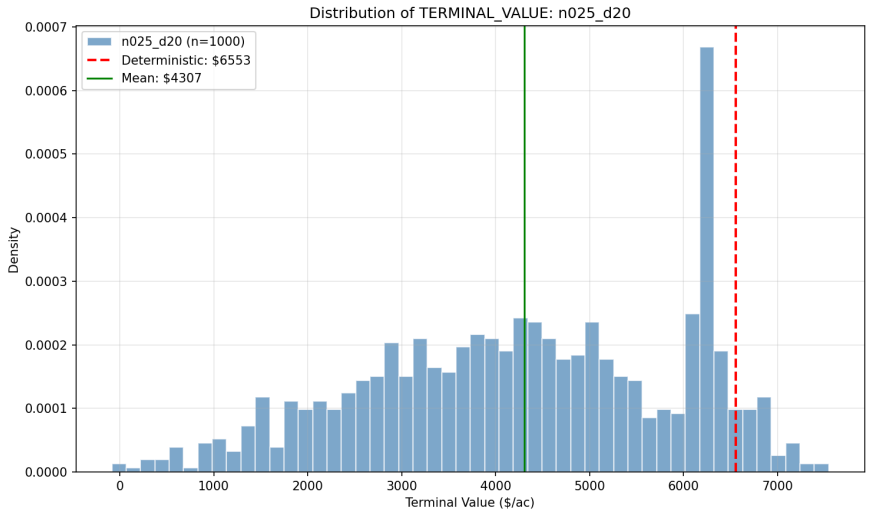

Figure 8: Terminal value distribution: `n025_d20` ($\lambda_{\text{proc}} = 0.25$, $p_{\text{dist}} = 1/20$).

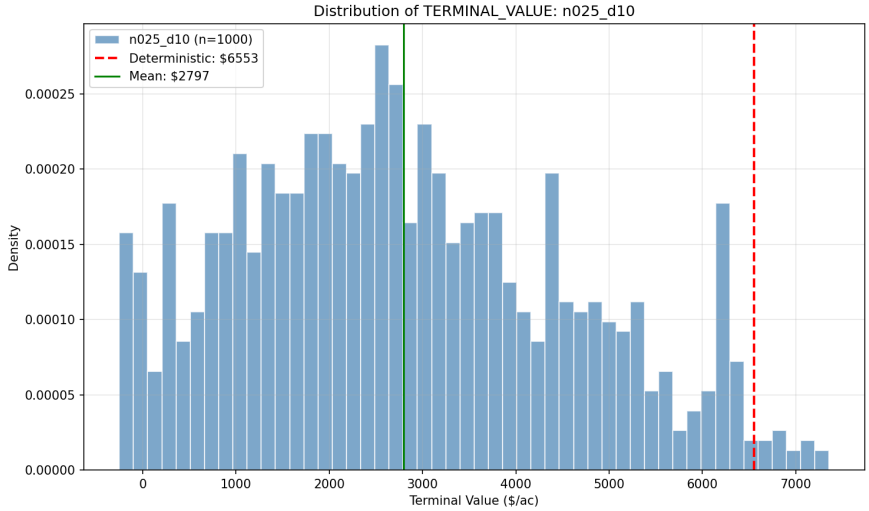

Figure 9: Terminal value distribution: `n025_d10` ($\lambda_{\text{proc}} = 0.25$, $p_{\text{dist}} = 1/10$).

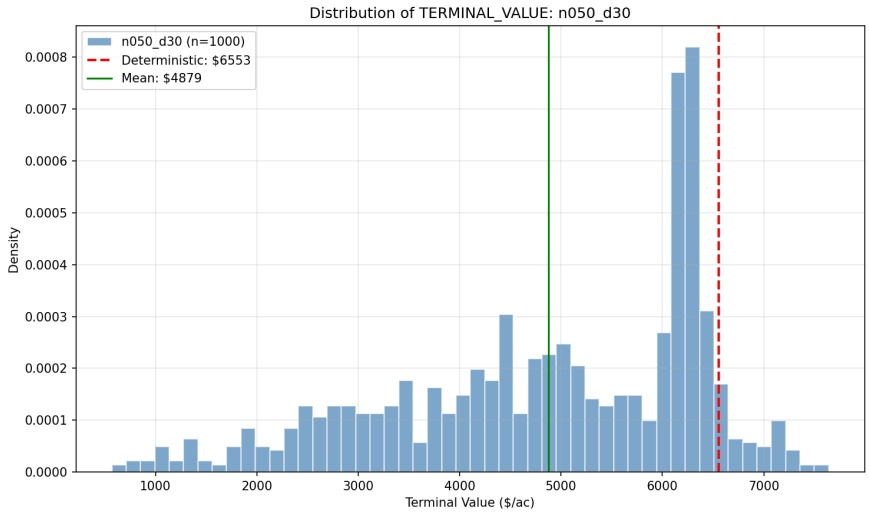

Figure 10: Terminal value distribution: `n050_d30` ($\lambda_{\text{proc}} = 0.50$, $p_{\text{dist}} = 1/30$).

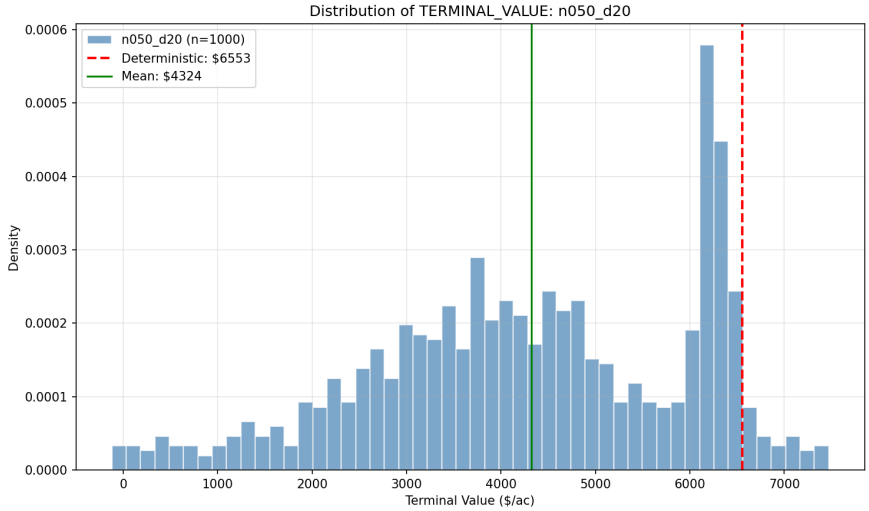

Figure 11: Terminal value distribution: `n050_d20` ($\lambda_{\text{proc}} = 0.50$, $p_{\text{dist}} = 1/20$).

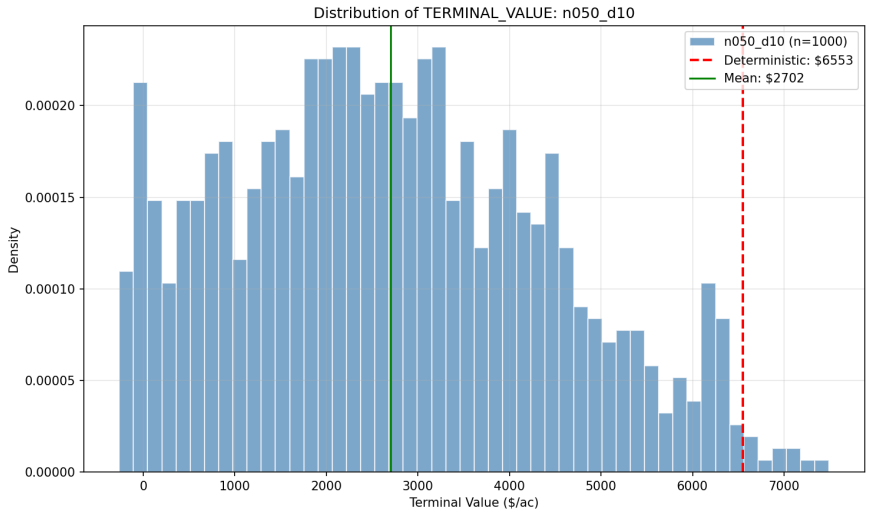

Figure 12: Terminal value distribution: `n050_d10` ($\lambda_{\text{proc}} = 0.50$, $p_{\text{dist}} = 1/10$).

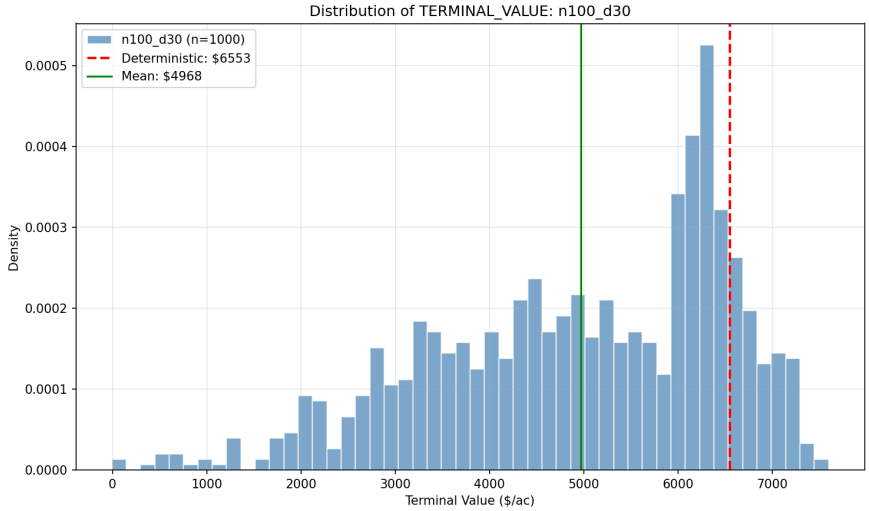

Figure 13: Terminal value distribution: `n100_d30` ($\lambda_{\text{proc}} = 1.00$, $p_{\text{dist}} = 1/30$).

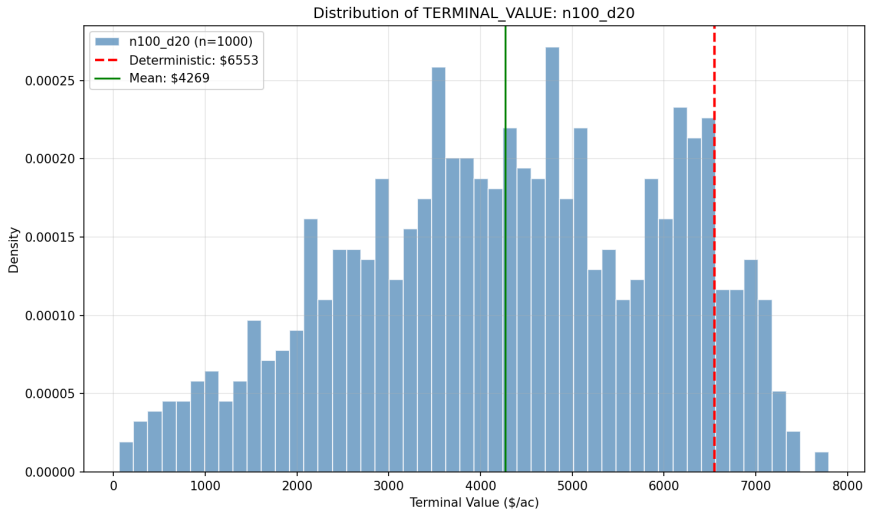

Figure 14: Terminal value distribution: `n100_d20` ($\lambda_{\text{proc}} = 1.00$, $p_{\text{dist}} = 1/20$).

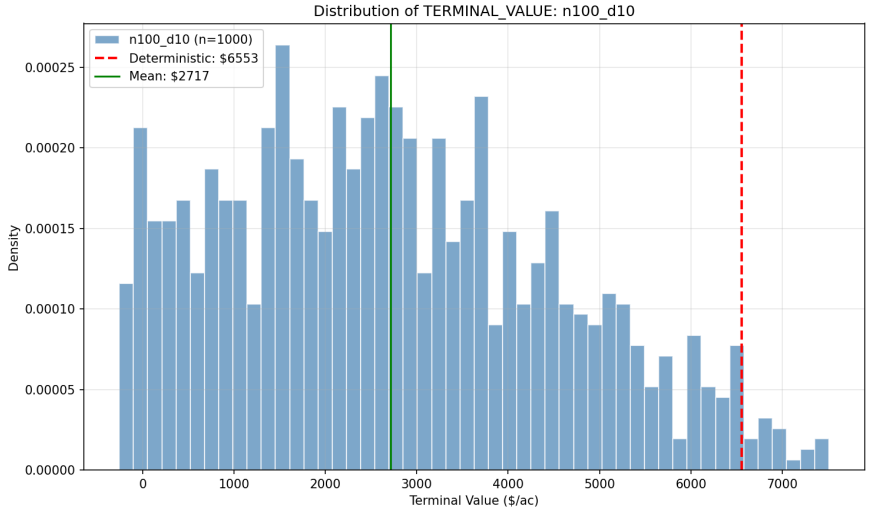

Figure 15: Terminal value distribution: `n100_d10` ($\lambda_{\text{proc}} = 1.00$, $p_{\text{dist}} = 1/10$).